SOUND

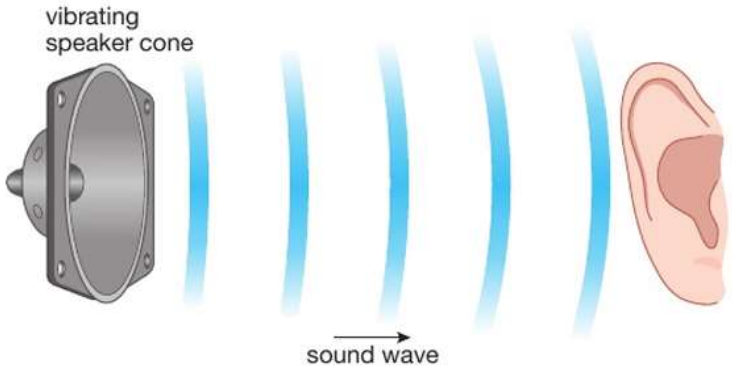
Learning Objectives

- Explain that sound waves are longitudinal waves which can be reflected and refracted.
- Classify the frequency range for human hearing and other animals
- Measuring or calculate the speed of sound in air
- Explain the application of sound
- Analyze the relation of pitch and frequency
- Analyze the relation of loudness and amplitude of a sound

Sound Waves

- sound waves are longitudinal waves which can be reflected and refracted.

Sounds are produced by objects that are vibrating. We hear sounds when these vibrations, travelling as sound waves, reach our ears.

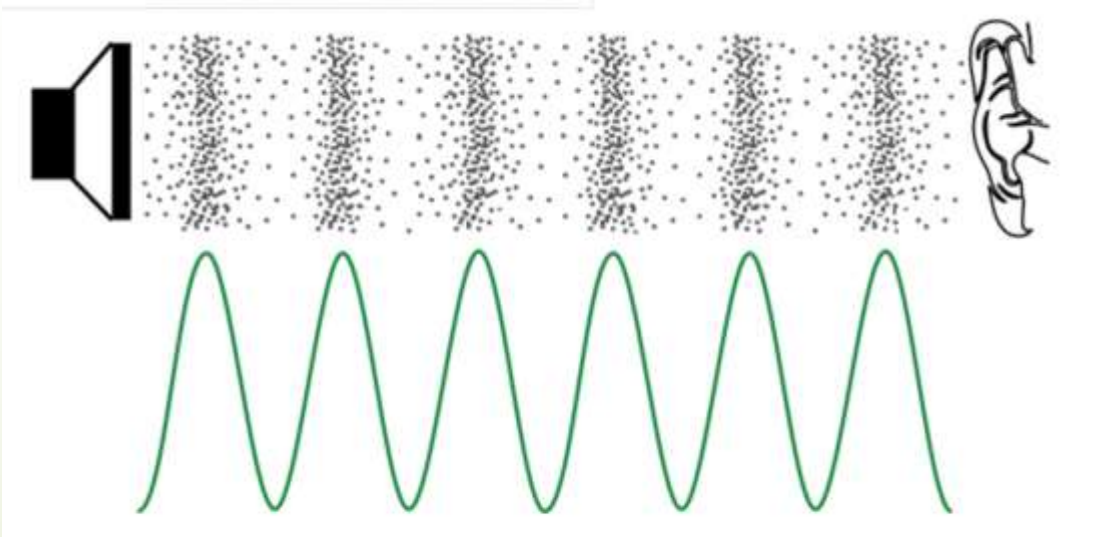


▲ Figure 13.3 The loudspeaker vibrates and produces sound waves.

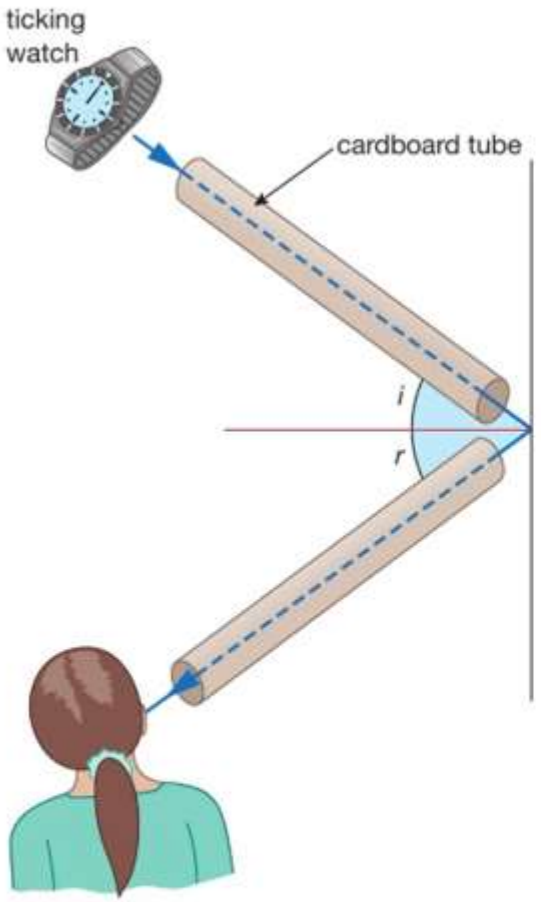
EXTENSION WORK

You will not be tested on this more detailed description of how sounds from a loudspeaker are heard. As the speaker cone moves to the right, it pushes air molecules closer together, creating a **compression**. These particles then push against neighbouring particles so that the compression appears to be moving to the right. Behind the compression as the speaker cone moves to the left is a region where the particles are spread out. This region is called a rarefaction. After the cone has vibrated several times, it has created a series of compressions and rarefactions travelling away from it. This is a longitudinal sound wave (see page 98). When the waves enter the ear, they strike the eardrum and make it vibrate. These vibrations are changed into electrical signals, which are then detected by the brain.

- Any terminologies: COMPRESSION (Rapatan) aknd RAREFACTION (Renggangan)



REFLECTION of SOUND WAVES



▲ Figure 13.4 Sound waves are reflected in the same way that light rays are reflected.

REFLECTION

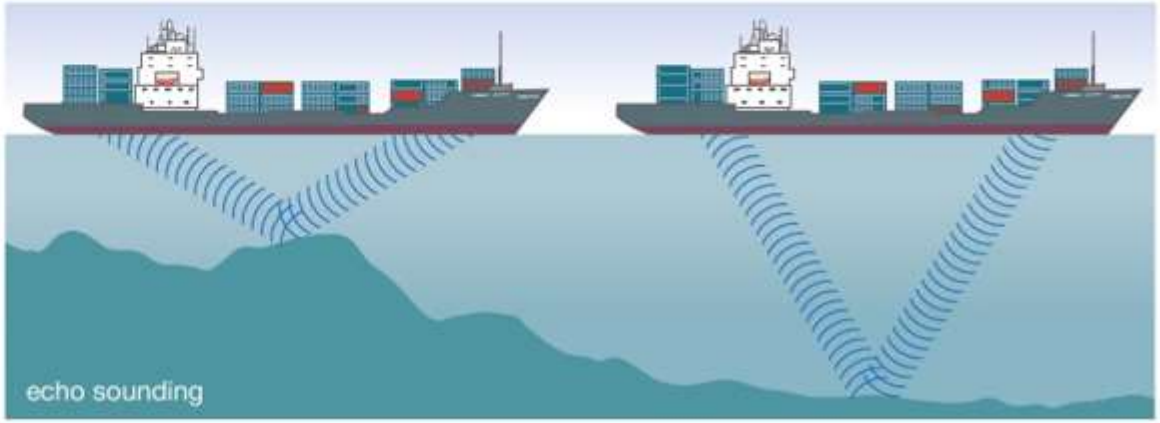
Sound waves behave in the same way as any other wave.

When a sound wave strikes a surface it may be reflected. Like light waves, sound waves are reflected from a surface so that the angle of incidence is equal to the angle of reflection (see Figure 13.4).

Ships often use echoes to discover the depth of the water beneath them. This is called echo sounding.

1. Sound waves are emitted from the ship and travel to the seabed (sea floor).
2. Some of these waves are reflected from the seabed back up to the ship.
3. Equipment on the ship detects these sound waves.
4. The time it takes the waves to make this journey is measured.
5. Knowing this time, the depth of the sea below the ship can be calculated.

The system of using echoes in this way is called **sonar** (Sound, Navigation And Ranging).

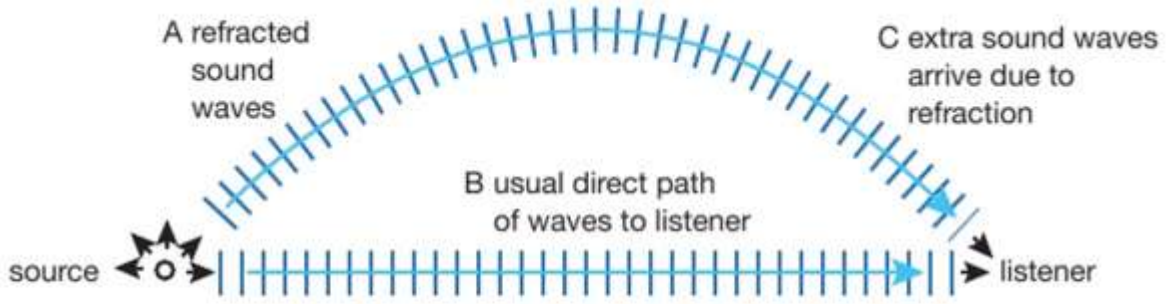


▲ Figure 13.5 Reflected sound can be used to tell ships about the depth of the sea beneath them.

REFRACTION of SOUND WAVES

All waves can be refracted, even sound waves! For example, if some parts of a sound wave are travelling through warm air, they will travel more quickly than those parts travelling through cooler air. As a result the direction of the sound wave will change. It will be refracted.

Although it is not possible to see sound waves being refracted, we can sometimes hear their effect. Standing at the edge of a large pond or lake we can sometimes hear sounds from things on the other side of the water much more clearly than we would expect. This is due to refraction. Figure 13.6 explains how this happens.



▲ Figure 13.6 Why sometimes sounds travelling across water are louder than we expect

1. Most of the sound we hear travels to us in a straight line (Path B).
2. But some sound travels upwards (Path A).
3. If the temperature conditions are right, then as the sound waves travel through the air they are refracted and follow a curved path downwards (Path C).
4. We now receive two sets of sound waves.
5. So the sound we hear seems louder and clearer.

Pitch and Frequency

Small objects, such as the strings of the violin in Figure 13.7, vibrate quickly and produce sound waves with a high frequency. These sounds are heard as notes with a high pitch.

Larger objects, such as the strings of the cello, vibrate more slowly and produce waves with a lower frequency. These sounds have a lower pitch.

The frequency of a source is the number of complete vibrations it makes each second. We measure frequency in hertz (Hz). If a source has a frequency of 50 Hz this means that it vibrates 50 times each second and therefore produces 50 waves each second.

Figure 13.7 The violin produces notes that are higher pitched than those from the cello.

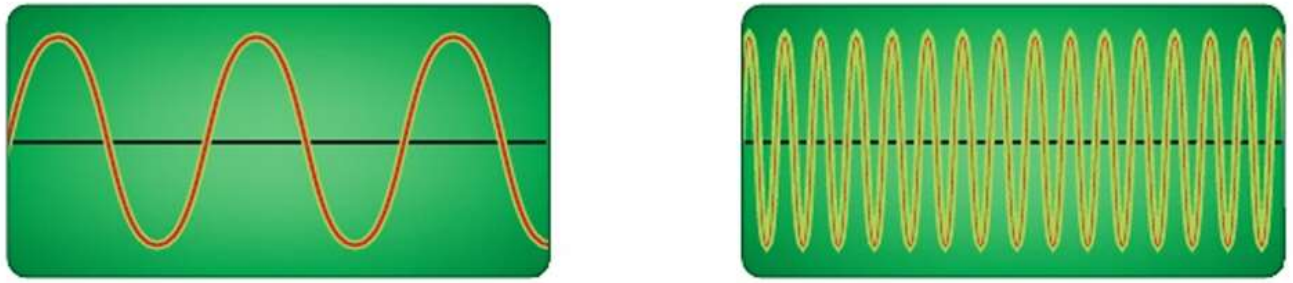


Figure 13.9 Low-pitched sounds and high-pitched sounds seen on an oscilloscope. A low-pitched sound has a low frequency, meaning fewer complete waves per second, so fewer complete waves are seen on the oscilloscope than for the high-pitched (high frequency) sound.

High pitch means high frequency

The loudness of sound

If the drum in Figure 13.12 is hit hard, lots of energy is transferred to it from the drum stick. The drum skin vibrates up and down with a large amplitude, creating sound waves with a large amplitude and we hear a loud sound. If the drum is hit more gently less energy is transferred and sound waves with a smaller amplitude are produced. We hear these as quieter sounds.

▲ Figure 13.13 Loud sounds and quiet sounds on an oscilloscope

▲ Figure 13.14 A scientist measuring the sound level as a car passes him

Larger the amplitude Louder the sound

Measuring the speed of sound in air

ACTIVITY 1

▼ PRACTICAL: INVESTIGATE THE SPEED OF SOUND USING ECHOES

You can measure the speed of sound using two bits of wood and a stopwatch. Stand some distance away from a friend, start the stopwatch when you see her bang the two bits of wood together and stop the stopwatch when you hear the sound. Now using the equation speed, $v = \frac{\text{distance, } s}{\text{time, } t}$ you can calculate the speed of sound. The answer you are likely to get will not be very accurate unless you have very good eyesight and very fast reactions.

You can obtain a more accurate measurement using echoes (sounds that are reflected off a surface).

1. Both you and your friend should stand at least 50 m away from a large wall or building.
2. Bang the two pieces of wood together and listen for the echo.
3. Bang the pieces of wood together each time you hear the echo. This will create a regular rhythm of claps.
4. Ask the friend to time you doing 20 claps. During this time the sound will have travelled, for example, 50 m × 20 × 2 (to the wall and back 20 times), and you can divide this distance by the time to work out the speed of the sound.

The speed of sound in air is approximately 340 m/s, although this value does vary a little with temperature.

EXAMPLE 1

A girl stands 100 m from a tall building clapping her hands each time she hears an echo of her clap. It takes 11.7 s for her to hear 20 echoes. Calculate the speed of sound.

$$v = \frac{s}{t}$$

$$= \frac{100\,\text{m} \times 2 \times 20}{11.7\,\text{s}}$$

$$= 342\ \text{m/s}$$

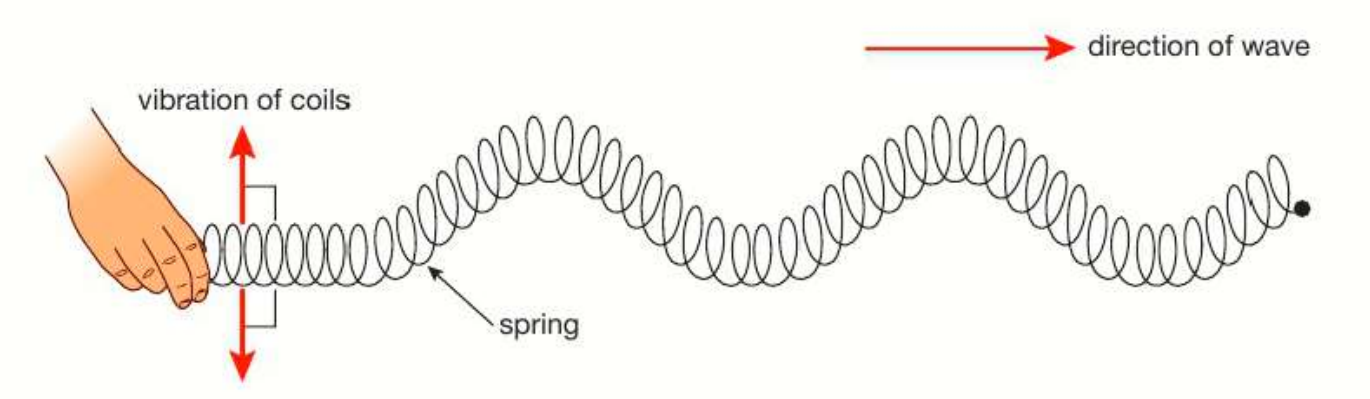
PROPERTIES OF WAVES

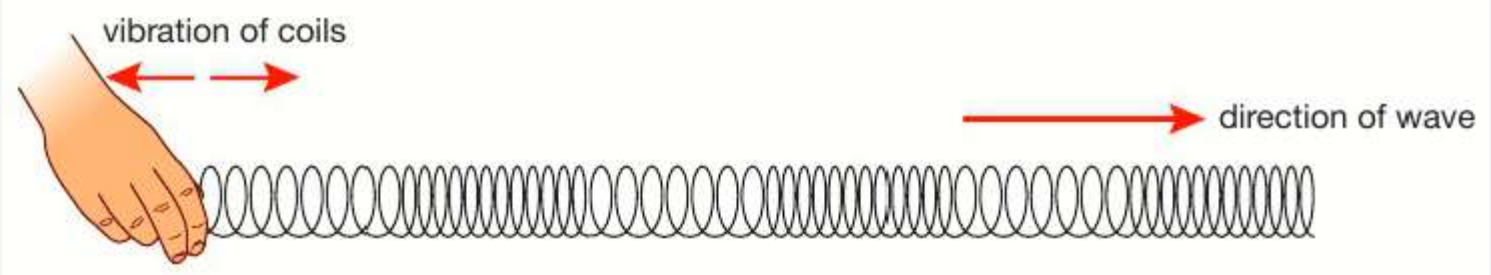
Learning Objectives:

1. *Explain the definition of a wave*
2. *Differ transverse & longitudinal waves*
3. *Define amplitude, wavelength, wavefront, period and frequency of a wave*
4. *Apply the relationship speed, frequency, and wavelength of a wave*
5. *Recall the relationship of frequency & period of a wave*

WAVES

Definition





A wave is a vibration that propagates in a certain direction.

A wave is transferring energy from place to place.

WAVEFRONT

Definition

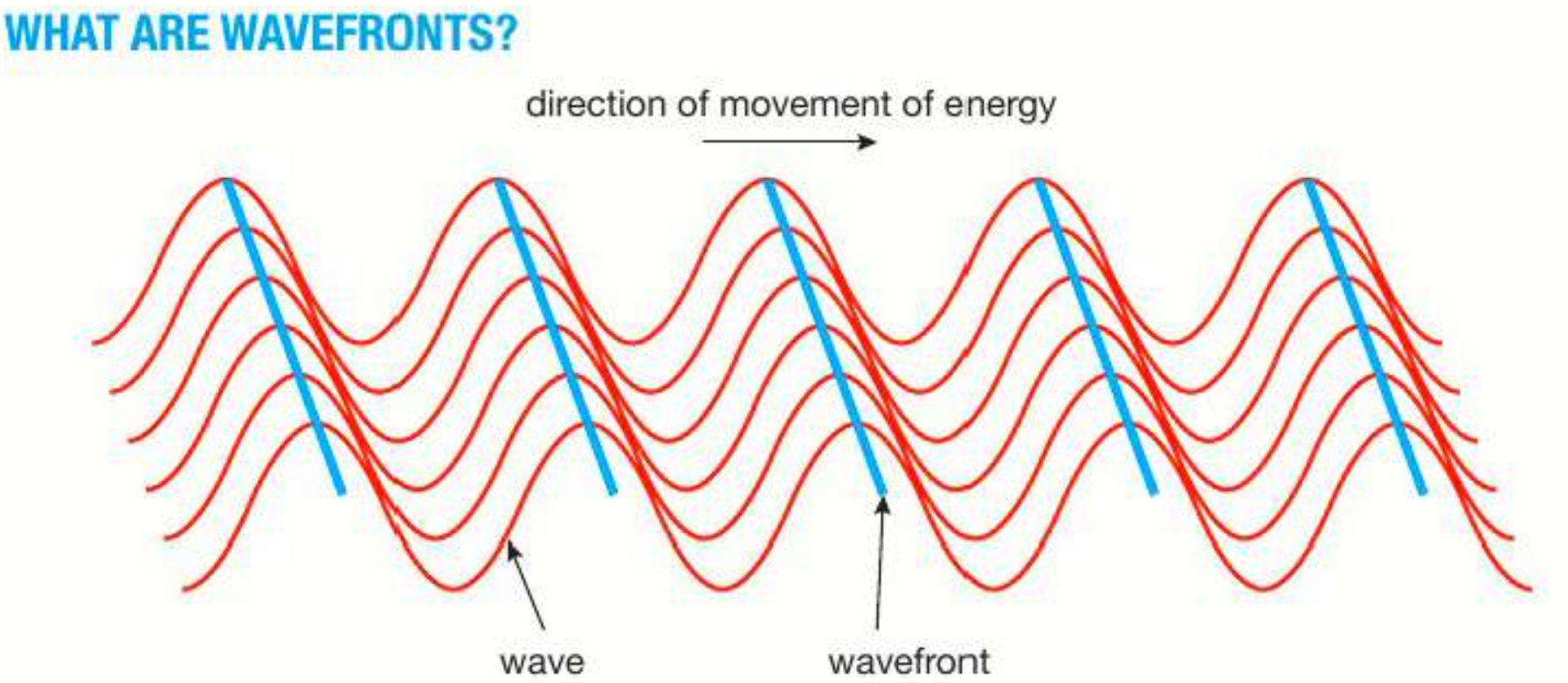
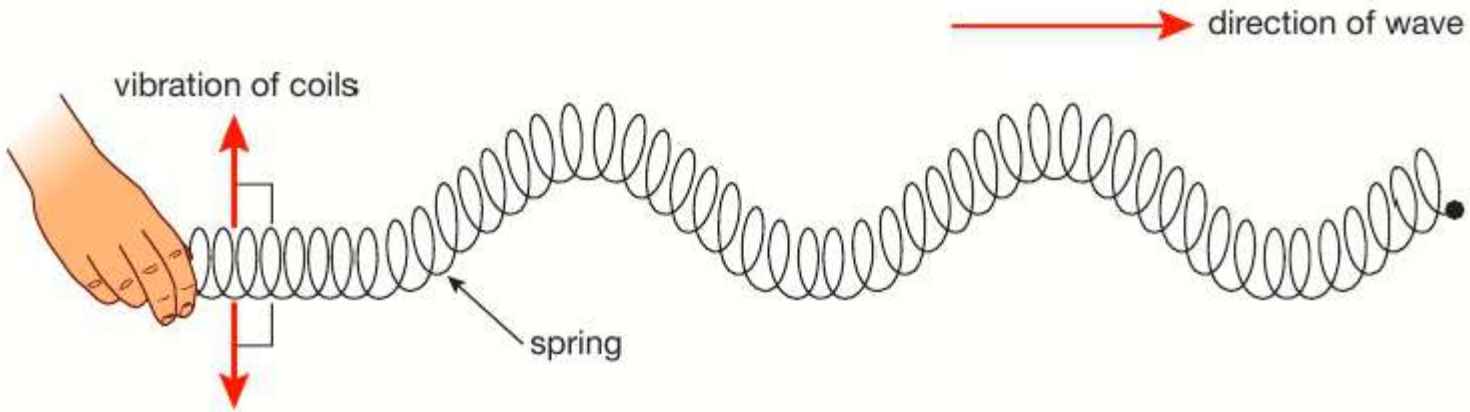


Figure 10.3 Wavefronts are created by overlapping lots of different waves. A wavefront is a line where all the vibrations are in phase and the same distance from the source.

Transverse Waves

Definition

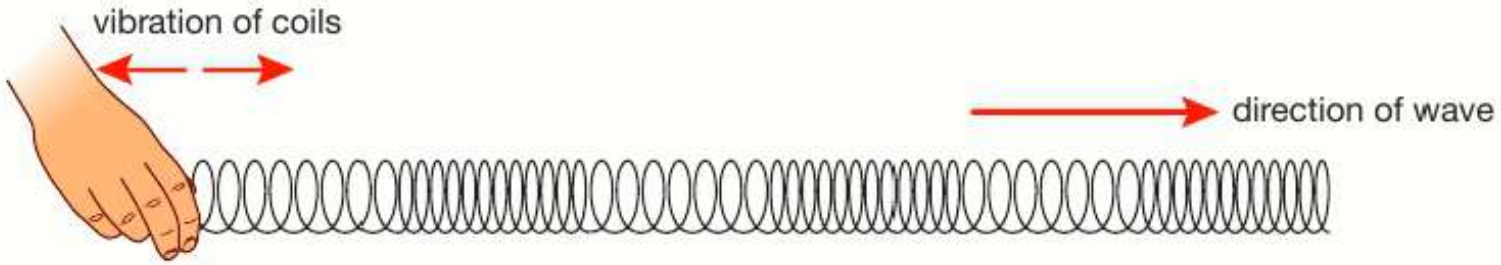
A transverse wave is one that vibrates, or oscillates, at right angles to the direction in which the energy or wave is moving. Examples of transverse waves include light waves and waves travelling on the surface of water.



▲ Figure 10.4 A transverse wave vibrates at right angles to the direction in which the wave is moving.

Longitudinal Waves

Definition



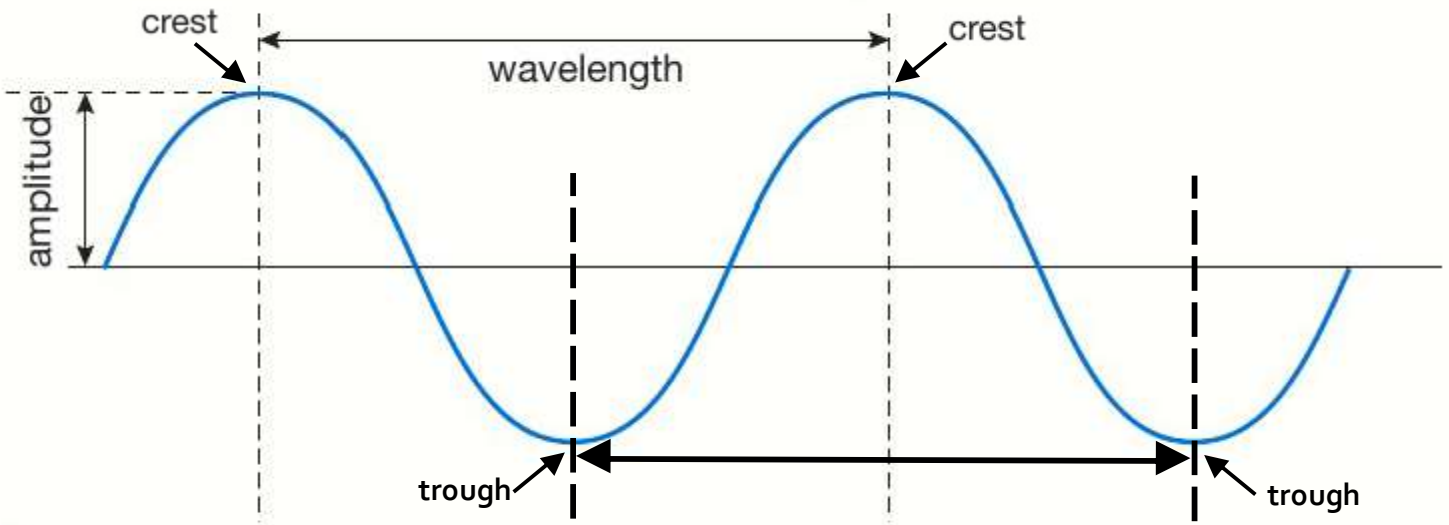
▲ Figure 10.5 A longitudinal wave vibrates along the direction in which the wave is travelling.

A longitudinal wave is one in which the vibrations, or oscillations, are along the direction in which the energy or wave is moving. Examples of longitudinal waves include sound waves.

Amplitude and Wavelength

Recognition

When a wave moves through a substance, its particles will move from their equilibrium (resting position). The maximum movement of particles from their resting or equilibrium position is called its **amplitude** (A).



▲ Figure 10.6 A wave has amplitude and wavelength.

The distance between a particular point on a wave and the same point on the next wave (for example, from crest to crest) is called the **wavelength** (λ).

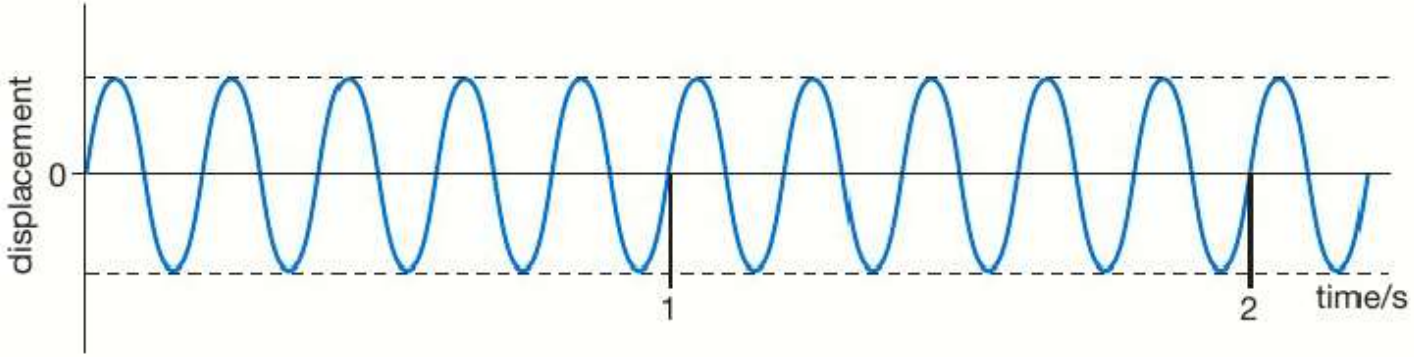
Or the wavelength can be the distance from trough to trough

KEY POINT

λ is the Greek letter lambda and is the usual symbol for wavelength.

Frequency & Period of a wave

Description



▲ Figure 10.8 This graph shows a wave with a frequency of 5 Hz.

The time it takes for a source to produce one wave is called the **time period** of the wave (T). It is related to the frequency (f) of a wave by the equation:

$$\text{frequency, } f\,(\text{Hz}) = \frac{1}{\text{time period, } T\,(\text{s})}$$

$$f = \frac{1}{T}$$

This equation can also be written as

$$T = \frac{1}{f}$$

EXAMPLE 1

Calculate the period of a wave with a frequency of 200 Hz.

$$T = \frac{1}{f}$$

$$= \frac{1}{200\ \text{Hz}}$$

$$= 0.005 \text{ or } 5\ \text{ms } (1000\ \text{ms} = 1\ \text{s})$$

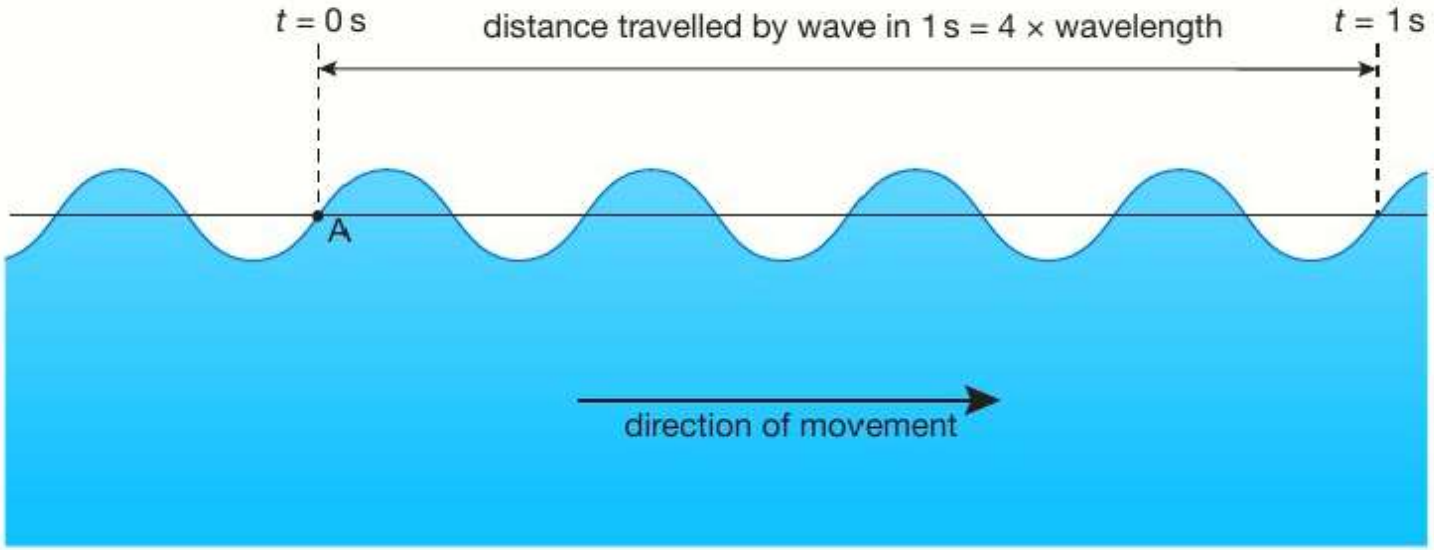
How to Apply the relationship speed, frequency, and wavelength of a wave

THE WAVE EQUATION

There is a relationship between the wavelength (λ), the frequency (f) and the wave speed (v) that is true for all waves:

wave speed, v (m/s) = frequency, f (Hz) × wavelength, λ (m)

$$v = f \times \lambda$$



▲ Figure 10.10 A wave with a frequency of 4 Hz

Imagine that you have created water waves with a frequency of 4 Hz. This means that four waves will pass a particular point each second. If the wavelength of the waves is 3 m, then the waves travel 12 m each second. The speed of the waves is therefore 12 m/s.

$$v = f \times \lambda$$
$$= 4\ \text{Hz} \times 3\ \text{m}$$
$$= 12\ \text{m/s}$$

How to Apply the relationship speed, frequency, and wavelength of a wave

EXAMPLE 2

A **tuning fork** creates sound waves with a frequency of 170 Hz. If the speed of sound in air is 340 m/s, calculate the wavelength of the sound waves.

$$v = f \times \lambda$$

So $$\lambda = \frac{v}{f}$$

$$= \frac{340 \text{ m/s}}{170 \text{ Hz}}$$

$$= 2 \text{ m}$$

▲ Figure 10.11 A tuning fork

Thank You

for your good attention